

Raj Sahoo

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EDUCATION

Parul University

B.Tech – Information Technology

- CGPA: 7.67/10.0

Vadodara, Gujarat

2021 – 2025

EXPERIENCE

AI/ML Software Developer

ConstructMind.ai, Vadodara

May 2026 – Present

- Used **LangGraph** to build a document-reading agent that autonomously generates full construction schedules, and a separate **multi-agent** chat system (CoCo) that classifies user intent and routes between a greeting agent, a read-only question-answering agent, and a schedule-editing agent
- Used **guardrails** (iteration/token budgets, repetition detection, drift detection, context-window preflight checks) to stop the agent from looping, stalling, or running unnecessary calls
- **Sandboxed** the agent's shell access with a command denylist, a scrubbed environment (no secrets/network binaries), and path-traversal guards so it could run open-ended Python safely
- Used **Anthropic strict-mode tool calling** to enforce schema-validated, structured tool inputs instead of parsing free-text model output
- Parsed and chunked project documents, converted them to **embeddings**, stored them in **pgvector**, and ran cosine-similarity **semantic search** to shortlist relevant content before LLM verification
- Paired deterministic engines (critical-path scheduling, WBS coding, DCMA-style validation) with the LLM only where judgment was genuinely needed, keeping math and scoring reproducible

AI/ML Software Developer

MyOnsite Healthcare, Vadodara

Dec 2024 – May 2026

- **Carevio – Production AI Chatbot:** Built a HIPAA-aligned LangGraph chatbot for a nationwide phlebotomy company, backed by ChromaDB (RAG), Redis, and MongoDB, with a semantic cache and smart gpt-4o-mini/o3-mini routing cutting response latency to 1-3s; Dockerized and deployed on AWS EC2
- **Voice AI Agent:** Built a real-time telephony agent on the GPT-4o Realtime API with Whisper transcription, ElevenLabs synthesis, and LangGraph orchestration for appointment scheduling and human-agent escalation
- **Sentiment Analysis Dashboard:** Automated pipeline transcribing and analyzing S3 call recordings (Whisper + GPT-4) into per-agent performance dashboards and daily AI-generated reports
- **CRM MCP Server:** Deployed a multi-tenant MCP server letting AI assistants query a CRM in plain English, with SELECT-only enforcement and bcrypt-authenticated access
- **AI Proctoring System:** Built a real-time proctoring service using YOLOv8n and MediaPipe Face Mesh/Detection for prohibited-object, head-turn, and multi-face violation detection
- Also built a ChromaDB + KNN lead-qualification pipeline and an n8n social-media automation workflow (Perplexity, WordPress, Freepik)

PROJECTS

Multimodal Image Captioning System | *Python, TensorFlow, CNN, LSTM, Attention Mechanisms*

- Built and trained a supervised CNN-LSTM model with attention mechanisms from scratch on a Kaggle dataset; evaluated caption quality using BLEU, METEOR, and CIDEr metrics

Real-Time Sign Language Recognition System | *Python, OpenCV, TensorFlow, MediaPipe*

- Collected and labeled a custom ASL dataset with data augmentation; trained a CNN from scratch and integrated MediaPipe hand landmark detection with OpenCV for real-time low-latency inference

Employee Promotion Prediction Engine | *Python, Scikit-learn, XGBoost, FastAPI*

- Led a team of 4 to build an HR analytics solution using ensemble methods (Logistic Regression, Random Forest, XGBoost) with SHAP-based interpretability; deployed with a FastAPI backend and interactive dashboard for HR teams

Plant Disease Classification System | *Python, TensorFlow, Keras, OpenCV, scikit-image, Pandas, Seaborn*

- Built and trained a custom 3-block CNN from scratch on 87,000+ leaf images across 38 disease categories (14 plant species), achieving 96.57% validation accuracy in 10 epochs using TensorFlow/Keras Sequential API with progressive filter scaling (32 → 64 → 128), dual Dropout regularization (0.2 + 0.5), and Adam + categorical crossentropy
- Engineered a memory-efficient data pipeline using `ImageDataGenerator.flow_from_directory` with pixel normalization, 256×256 resizing, and batch loading (batch size 32) to handle 70K training images without loading the full dataset into RAM
- Evaluated with per-class classification report and 38×38 seaborn confusion matrix; macro-average precision, recall, and F1 all at 96% across all disease categories on a 17,572-image held-out validation set

TECHNICAL SKILLS

LLM & Agentic AI: LangGraph, LangChain, RAG, LLMs, Multi-Agent Systems, Anthropic Claude Tool Use (Strict Mode), Prompt Engineering, LLM-as-Judge, MCP Servers

AI Frameworks & Tools: Agent Guardrailing, Sandboxing, OpenAI API, LlamaParse, Langfuse, ChromaDB, pgvector, Embeddings, Semantic Search, Sentence Transformers

Machine Learning & Computer Vision: Python, NLP, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, CNN, LSTM, YOLO, MediaPipe, XGBoost, Hugging Face, Feature Engineering

Backend & APIs: FastAPI, Flask, REST APIs, Socket.IO, WebSockets, n8n Automation, BullMQ

Databases, Docker & Infrastructure: Docker, Git, GitLab, MongoDB, Redis, PostgreSQL, MySQL, AWS (EC2, S3, Lambda, RDS, CloudWatch), Azure Blob Storage

ACCOMPLISHMENTS & LEADERSHIP

AI/ML Hackathon 2024 – Participated in Parul University's AI/ML hackathon conducted by myOnsite Healthcare

Shaastra Programming Contest 2023 – Competed at IIT Madras in national-level algorithmic programming competition

Technical Mentorship – Served as judge and mentor at an AI/ML hackathon organized by myOnsite Healthcare; assessed project implementations and scored teams on technical execution and innovation

Team Leadership – Mentored and oversaw 10 interns at myOnsite Healthcare building enterprise-grade AI/ML systems; conducted code reviews, established standards, and ran knowledge transfer sessions

Production Impact – Systems in active production: ConstructMind.ai's multi-agent scheduling platform and COCO editing agent; myOnsite's Carevio chatbot, Voice AI agent, CRM MCP server, sentiment analysis dashboard, and AI proctoring platform handling real customer traffic across multiple departments